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Movie Recommendation System Using Hybrid Filtering

Introduction

The Movie Recommendation System suggests unrated films by combining rating patterns, movie genres, and overall popularity. This project uses the MovieLens latest-small dataset: 100,836 ratings for 9,742 movies from 610 anonymized users.

It demonstrates a reproducible recommendation workflow for an educational setting. MovieLens IDs do not provide user demographics, and recommendations are examples rather than evidence of individual preferences.

Problem Statement

Collaborative filtering estimates a user's interest from rating deviations among the 20 nearest users; content-based filtering measures genre overlap with highly rated movies. The hybrid rank combines normalized collaborative preference (60%), genre similarity (30%), and smoothed popularity (10%).

Personalized lists exclude titles already rated by the selected user. The project compares rating-prediction error against a global-mean baseline and checks top-10 recommendations against held-out ratings of at least four stars.

Results and Discussion

A per-user random 80/20 split (seed 42) used 80,672 ratings for training and held out 20,164. User-user collaborative filtering achieved RMSE 0.953 and MAE 0.728, compared with 1.044 and 0.827 for the global-mean baseline. The next page also summarizes ranking results.

Offline Evaluation

The per-user random 80/20 holdout contains 80,672 training ratings and 20,164 test ratings (seed 42). The table compares rating-prediction errors; lower values are better.

Model	RMSE	MAE
Global-mean baseline	1.044	0.827
User-user collaborative filtering	0.953	0.728

For anonymized user 1, leading hybrid results were Three Kings (1999) (0.904), City of God (Cidade de Deus) (2002) (0.903), and The Lego Movie (2014) (0.902).

Recommendation Code

```
movies, ratings = load_dataset()
recommender = MovieRecommendationSystem(movies, ratings)
recommendations = recommender.recommend_for_user(user_id=1, method='hybrid', top_n=10)


Hybrid ranking (top 10; held-out like ≥ 4 stars; 604 users):
Precision@10 0.027  |  Recall@10 0.022  |  Hit Rate@10 0.205
```

Collaborative rating estimates reduce RMSE by 0.091 and MAE by 0.099 against the global-mean baseline. The hybrid top-10 metrics are modest and depend on one random holdout. These offline measures do not show whether people would like or use the recommendations.

Conclusion

This project combines user-user collaborative filtering, genre similarity, and smoothed popularity in a reproducible movie recommender. On this random holdout, collaborative estimates improve over the global-mean rating baseline, while top-10 retrieval remains limited. MovieLens latest-small is a development sample; the results are not a user study or a claim about real-world satisfaction.

Reference

Harper, F. M., & Konstan, J. A. (2015). The MovieLens Datasets: History and Context. ACM Transactions on Interactive Intelligent Systems, 5(4), 19:1–19:19. https://doi.org/10.1145/2827872. Dataset: MovieLens latest-small, GroupLens Research, https://grouplens.org/datasets/movielens/latest/ (non-commercial educational use).